

HANISHA VEMIREDDY

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PROFESSIONAL EXPERIENCE

Data Scientist II | Corteva Agriscience, Indianapolis, IN

Aug 2022 – Present

- Characterizing the challenges to improve the ability of smaller scale experiments to predict performance of similar conditions at larger scales by building our own ETL and data pipelines and making them more model-ready
- Design regular hypothesis testing, gather data, perform statistical analysis, and support research through simulations, linear mixed modeling, and statistical analyses to improve the effectiveness of research experiments to increase productivity of a crop protection product

Graduate Assistant | Iowa State University, Ames, IA

Aug 2017 – Aug 2022

Research Assistant

Constructing training data using two-way table of means: An application in Plant Breeding

- Constructed training data method using observations from similar genotypes, as an alternative to aggregating observations from each target genotype distribution, improving prediction accuracy by 40% on an average.
- Implemented different prediction models like OLS, CART, Random Forest, XG Boost, Neural Network models and to show that the new training data improved the prediction accuracy of yield.

A Regression approach to identify Discriminating Locations

- Proposed a novel method that can be applied to past multi-environment data to identify which test location best discriminate the genotype main effects, with a 30% improvement in the correct advancement decisions made in the early-stage experiments.
- Defined indices to characterize environmental harshness using phenotype data of genotypes.

Quantifying Uncertainty of Rank in Plant Breeding Experiment

- Captured the uncertainty of ranks of genotypes due to the specific set of environments observed by proposing a bootstrapping approach for constructing confidence intervals around rank
- Showed that the empirical coverage of the confidence interval closely matches the theoretical coverage of ranks, even when the data is up-to 95% sparsity

Teaching Assistant

- TA for the course “Stochastic Modelling, Analysis and Simulation” with 130 undergraduate students.

Data & Operations Research Intern | Principal Financial Group, Des Moines, IA

July 2019 – Dec 2019

Quality check on market sentiments obtained from a third-party service.

- Implemented an automated process to screen financial news data to help investment decisions using Python (Matplotlib, Plotly, Pandas, and Scikit-Learn), managing data with SQL (Snowflake).
- Defined different metrics for testing correlations, sparsity and periodicity at both market and company levels.

- Automated the entire pipeline of screening and validating the results of the algorithm on Amazon AWS platform and documented the whole pipeline using Sphinx.
- Demonstrated results in a Power BI live dashboard to the stakeholders regularly.

Consultant, Data Science | Widasconcepts Pvt Ltd, Bangalore, India Aug 2016– June 2017

- Implemented a module to determine the perimeter search for POIs (Point of Interests, e.g., gas stations) as the user travels, which reduced the module outsourcing costs to the company.
- Implemented logic for a Bluetooth-based Vehicle Identification module that automated the process of logbook entries.
- Designed a rule-based algorithm to predict user's home and office locations using real-time GPS trackers to improve the customer experience.

ACADEMIC PROJECTS

Fraud Detection at Self-Checkouts in Retail | Iowa State University Aug 2019 – May 2019

Guide: Dr. Stephen B Vardeman

- Built an ensemble model for classification of frauds using data from self-check counters at an established food retailer as part of Data Mining Cup 2019.
- Placed top 10 teams among the 150 teams participated in the competition worldwide.

Minimizing Tracking Error Under Uncertainty | Iowa State University Jan 2019 – May 2019

Guide: Dr. Sarah Ryan

- Designed a novel non-linear optimization algorithm to match the statistical moments of the historical stock return distribution to generate discrete future stock scenarios to represent stock market uncertainty using R & Julia. Presented results using R Shiny application.
- Built a stochastic mixed-integer linear program to mimic and minimize the tracking error between the returns of the portfolio and market benchmark under future stock return uncertainty. The tracking error is proven to be less than 5%.

EDUCATION

Ph.D., Industrial Engineering | Iowa State University (GPA 3.91) Aug 2022

Major Courses: Analysis of Stochastic Systems, Multivariate Modelling, Machine Learning, Integer Programming and Applied Time Series

Master of Science in Quality Management Science | Indian Statistical Institute June 2016

Major Courses: Non-Integer Programming, Design of Experiments, Linear Programming and Game Theory

Bachelor of Mathematics (Hons.) | Indian Statistical Institute June 2014

Major Courses: Linear Algebra, Optimization, Probability and Statistics

SKILLS

Languages: R, Python, Matlab, SQL

Optimization Tools: Gusek, Lingo, Octave, Julia

Data Science and Machine Learning: Caret, Dplyr, Tidyverse, Pandas, Sklearn, lme4

Data Visualization Tools: R Shiny, Tableau, GGplot, Spotfire

Developer Tools: Git, Eclipse, Visual Studio, PyCharm

Cloud Technologies: AWS (EC2, S3)

Data Warehousing Tools: Matillion ETL, Snowflake SQL